



Spices Detection in Warehouses

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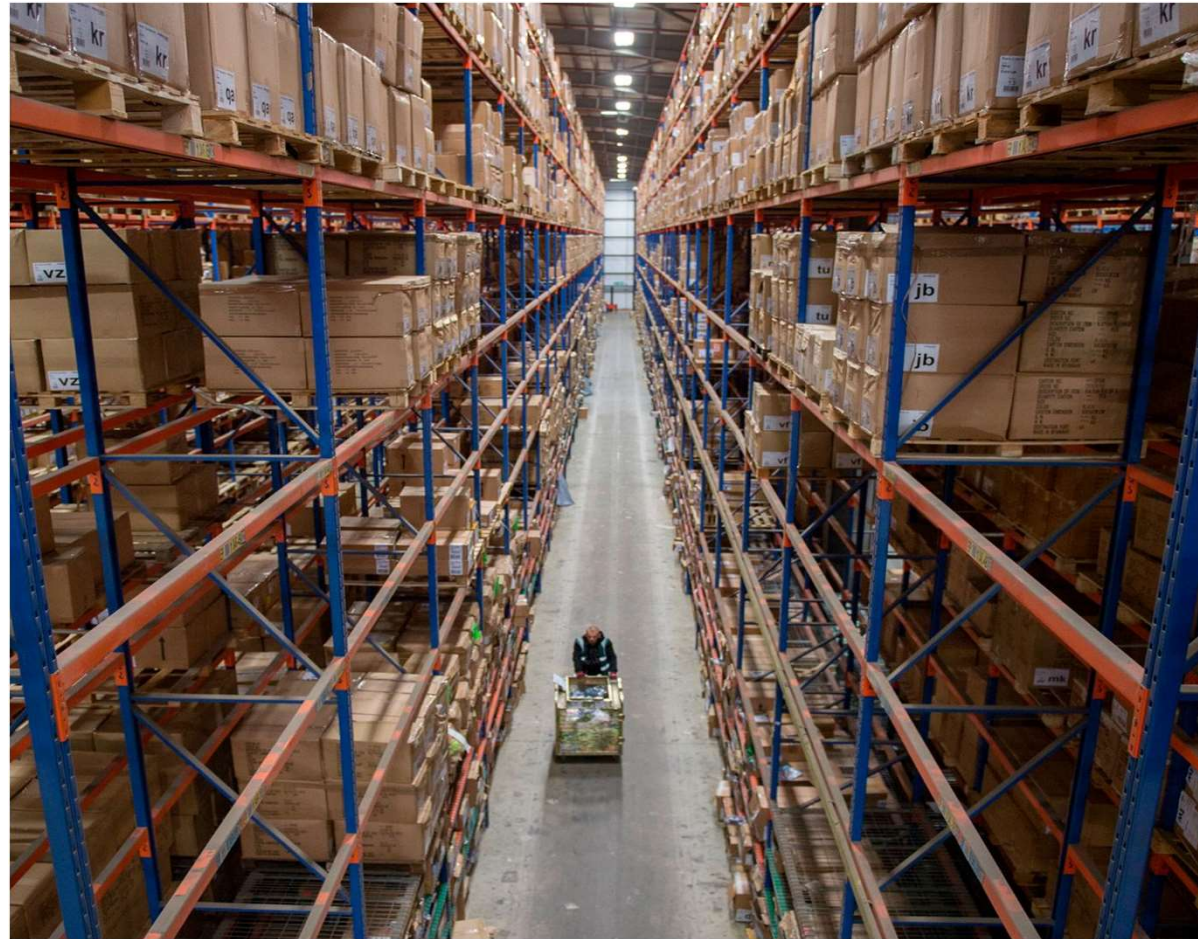


Deployment

Business Understanding

Warehouse Automation by eCommerce giants such as Amazon, eBay.

Major Problems- High labour costs, inaccurate inventory, redundant processes



Data understanding

- Choosing a list of spices is difficult as most of them are very small and in powders.
- Since powders are difficult to detect, packets and boxes containing those spices are used.
- Created datasets on own



Data Preparation

- A vast amount of time is spent on this phase which includes data collection, labelling and preprocessing.
- spices of smaller size and similar colours are not preferred to choose.



Dataset Collection

turmeric



cardamom



turmeric



turmeric



bayleaf



bayleaf



turmeric

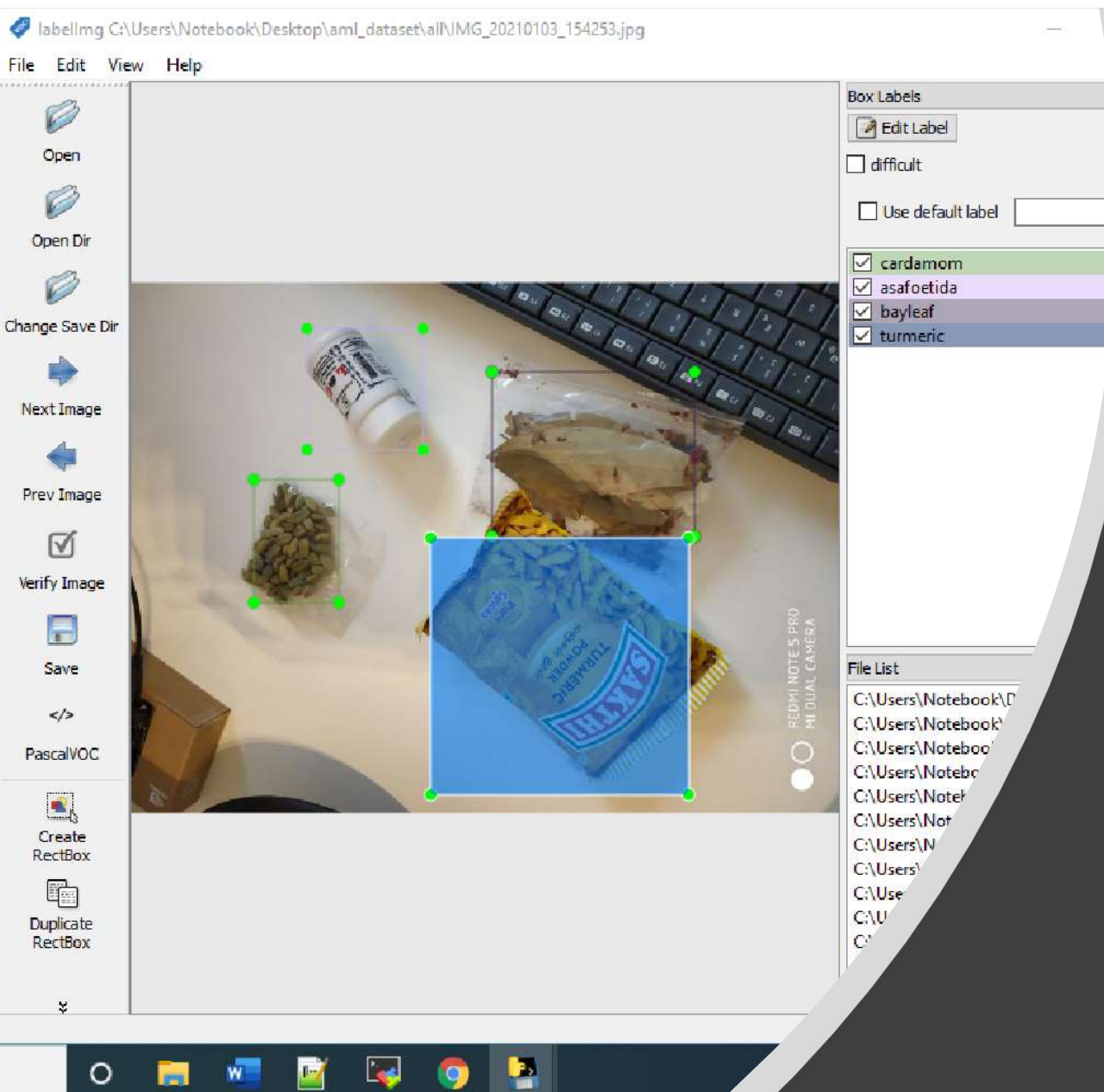


turmeric



bayleaf





1.Dataset
Labelling

2.xml→csv

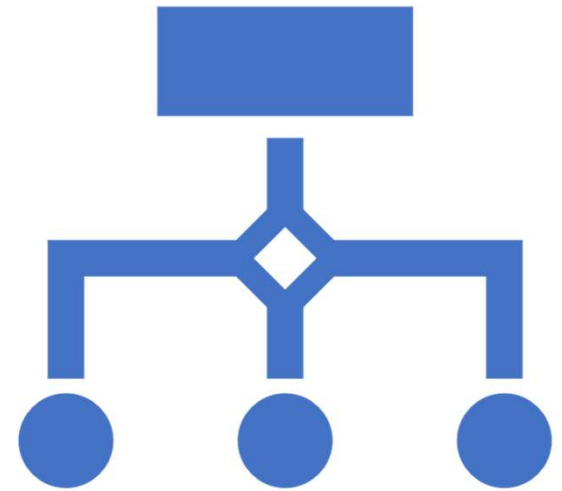


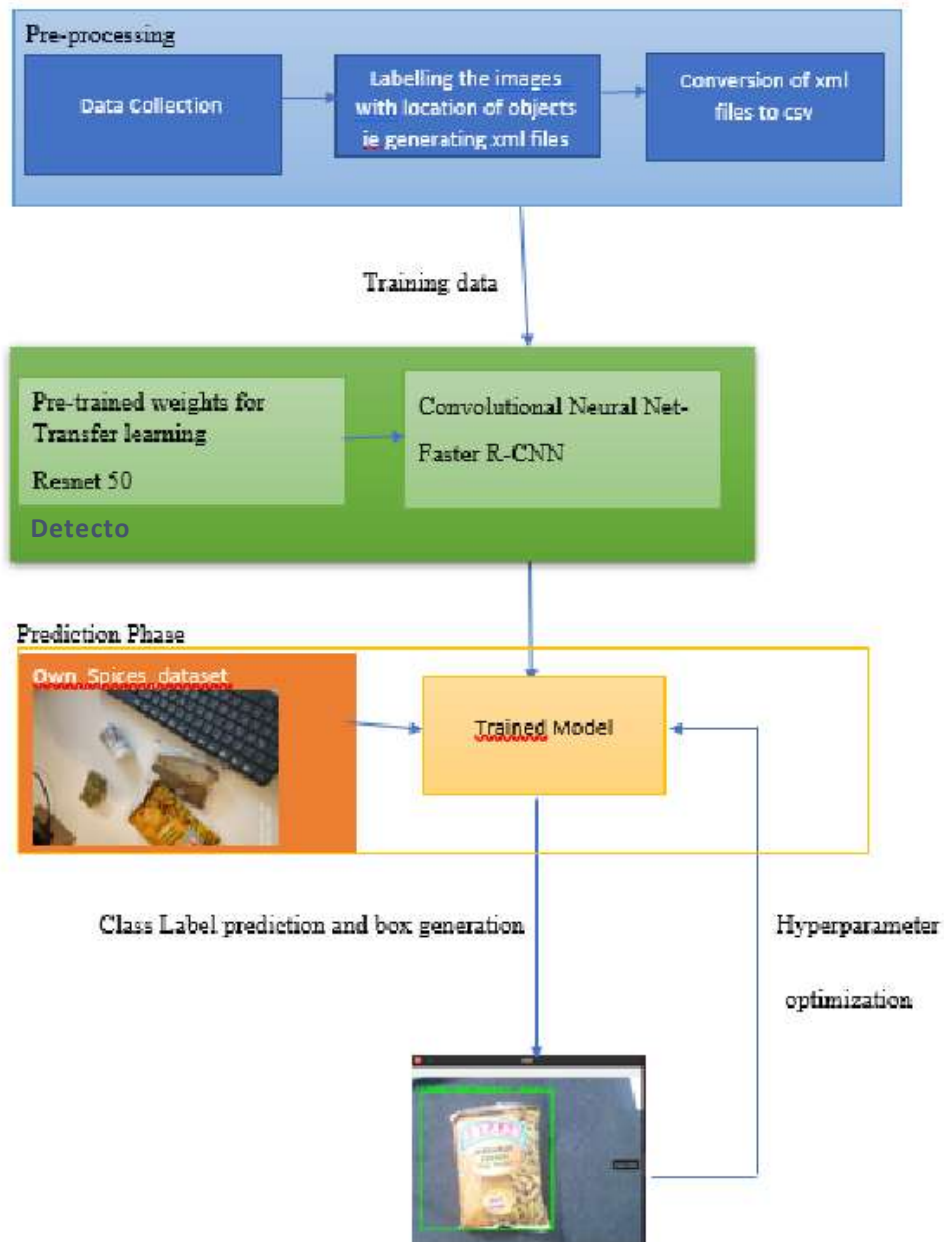
Data-Preprocessing

- Rescaling
- Colour adjustments
- normalization

Modeling- Initial Classification

- Data Augmentation- Flipping(horizontal), Rotation
- Pre-processing- inbuilt function from keras
- Pre-trained model Mobilnet v2
- Added a pooling layer to existing model
- Dense layer (activation- softmax)
- Dropout-0.2 (to prevent overfitting)
- Optimizer 'Adam' used for model compiling
- Loss- categorical cross entropy since more than two classes





Modeling

Evaluation- Classification Report

	precision	recall	f1-score	support
0	0.50	0.50	0.50	4
1	0.60	0.75	0.67	4
2	0.67	0.50	0.57	4
3	1.00	1.00	1.00	4
accuracy			0.69	16
macro avg	0.69	0.69	0.68	16
weighted avg	0.69	0.69	0.68	16



Spices Detection- Evaluation

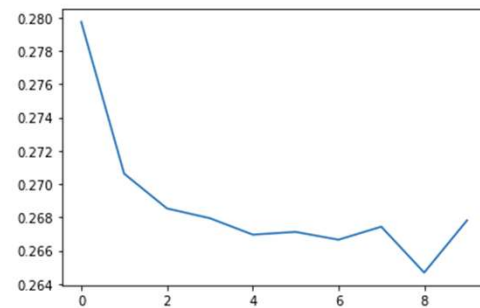
Spice detection- loss Analysis

```
plt.plot(losses)
plt.show()
model.save('new_spices_model_weight_jan11.pth')
```

begin iterating over validation dataset

100% |██████████| 72/72 [04:21<00:00, 3.63s/it]

Loss: 0.26782013869120014



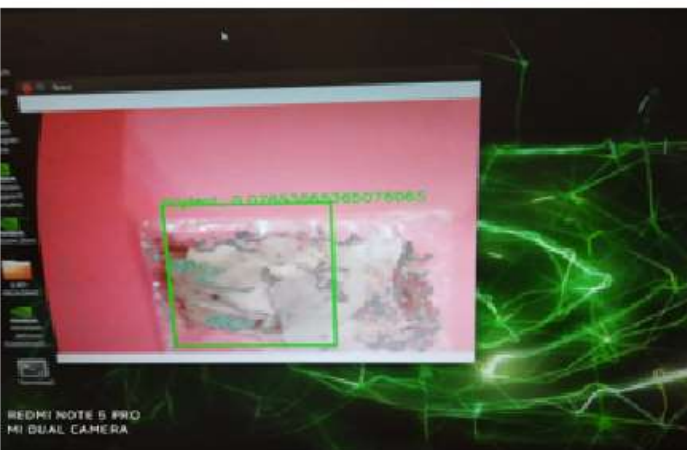
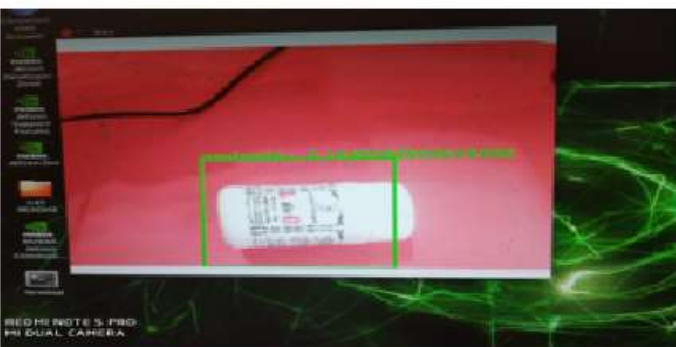
Integrating- camera

```
print("model loading success")
imcap = cv2.VideoCapture(2)

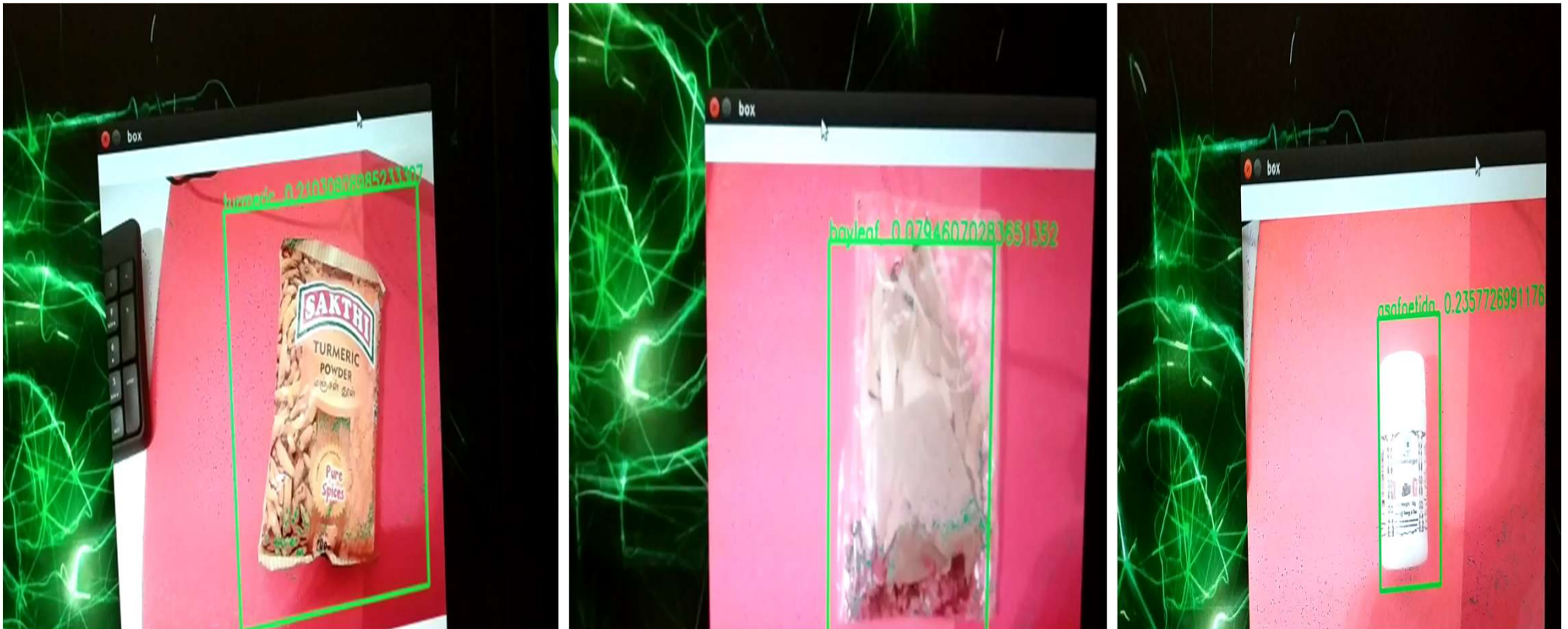
print("*****open="+str(imcap.isOpened()))
imcap.set(3, 640)
imcap.set(4, 480)

while True:
    for i in range(5):
        imcap.grab()
        success, image = imcap.retrieve()
        if success:
            #print("image read success")
            labels, boxes, scores = model.predict(image)
            if len(labels) > 0:
                for index, label in enumerate(labels):
                    if index < maxLabels:
                        coordinates=boxes[index].numpy()
                        startX = int(coordinates[0])
                        startY = int(coordinates[1])
                        endX = int(coordinates[2])
                        endY = int(coordinates[3])
                        probability = scores[index]
                        textValue = label+", "+str(float(probability))
                        image = cv2.rectangle(image,(startX, startY), (endX, endY), (0,255, 0), 3)
                        image = cv2.putText(image,textValue, (startX, startY), cv2.FONT_HERSHEY_SIMPLEX, 0.75, (0, 255, 0), 2)
                    else:
                        image = cv2.putText(image,"No Object Detected", (250,200), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
            cv2.imshow('box', image)
        else:
            print("Image loading failed")
            if cv2.waitKey(10) & 0xFF == ord('q'):
                break
    imcap.release()
cv2.destroyAllWindows('box')
```

```
model loading success
*****open=True
```



Spices Detection on Nano



Spices Detection with Nano on Real Time(Deployment)

Conclusion

- The results look only little promising provided with the extremely small set of training samples.
 - It can be improved further for sure if more time is spent.
- Detection of Indian spices is little challenging because of their smaller object size.



Thank You

